# Conditional probability and independence

## Lecture and worked examples

Maths Explorer

## Chapter path

**Objectives.** Change the reference sample space; build a tree; use a partition; distinguish disjoint events from independent events; handle up to four Bernoulli trials.

**Prerequisites.** Fractions, proportions, probabilities on a finite sample space, intersection and union.

**Yearly reference point.** 17-26 November: 8 hours

## Conditioning: changing the denominator

For $P(A) > 0$, the probability of B given A is $P_A(B) = \dfrac{P(A \cap B)}{P(A)}$. In a table of counts based on equally likely outcomes, it is the count of $A \cap B$ divided by the count of A. In general $P_A(B) \neq P_B(A)$.

## Conditioning: changing the denominator

Out of 100 people, 40 belong to A, 30 to B and 12 to both. $P_A(B) = \frac{12}{40} = 0.3$, whereas $P_B(A) = \frac{12}{30} = 0.4$. $P(A \cap B) = 0.12$.

## Conditioning: changing the denominator

“Among the A’s” fixes the denominator. A conditional probability is not an intersection.

## Conditioning: changing the denominator

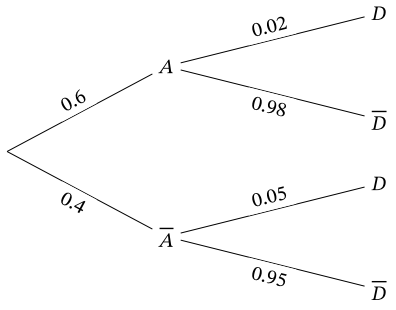
Out of 60 objects, 20 are red; among the red ones, 5 are damaged. Calculate $P(R \cap D)$ and $P_R(D)$.

## Trees, partitions and total probability

A partition $A_1, \ldots, A_k$ is a family of pairwise disjoint events whose union is the sample space. If their probabilities are positive, $P(B) = \sum_i P(A_i)P_{A_i}(B)$. On a tree, the probabilities leaving the same node sum to 1; the probabilities along a path are multiplied; disjoint paths are combined by adding.

# Trees, partitions and total probability

Two machines supply 60% and 40% of a production; their defect rates are 2% and 5%.
$P(D) = 0.6 \times 0.02 + 0.4 \times 0.05 = 0.032$. Among the defective items, the first machine accounts for $\dfrac{0.012}{0.032} = \dfrac{3}{8}$.



## Trees, partitions and total probability

The simple average of the rates is correct only if the branches have equal weights.

## Trees, partitions and total probability

$P(A) = \frac{1}{3}$, $P_A(B) = \frac{3}{4}$, $P_{\overline{A}}(B) = \frac{1}{4}$. Calculate $P(B)$.

## Independence and disjointness

A and B are independent if $P(A \cap B) = P(A)P(B)$. If $P(A) > 0$, this is equivalent to $P_A(B) = P(B)$. They are disjoint if $A \cap B = \emptyset$. Two disjoint events of strictly positive probability are not independent.

## Independence and disjointness

With a fair die, let A: the result is even, B: the result is at least 5. $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$, $P(A \cap B) = \frac{1}{6}$: independent. A and “the result is odd” are disjoint but not independent.

## Independence and disjointness

Independence in a model must be justified or stated explicitly; “two different events” is not enough.

## Independence and disjointness

Given $P(A) = 0.4$, $P(B) = 0.5$, $P(A \cap B) = 0.18$. Are they independent?

## Successive trials and Bernoulli trials

A Bernoulli trial has two outcomes: success, with probability p, and failure, with probability $1 - p$. In a repetition of n independent, identical trials, with $n \leq 4$ here, the branch probabilities remain p and $1 - p$. A path with k successes has probability $p^k(1 - p)^{n-k}$; we count the paths that realise the event.

## Successive trials and Bernoulli trials

For three trials with $p = \frac{1}{2}$, exactly two successes correspond to SSF, SFS, FSS. Each path has probability $\frac{1}{8}$, so the total is $\frac{3}{8}$. At least one success has probability $1 - (\frac{1}{2})^3 = \frac{7}{8}$.

## Successive trials and Bernoulli trials

When drawing without replacement from an urn, the probabilities generally change after the first draw: the independent model may be wrong.

## Successive trials and Bernoulli trials

Four independent trials, each with probability $p = \frac{1}{3}$. Calculate the probability of four failures, then of at least one success.

## Simulating and discussing the model

A pseudo-random simulation produces frequencies that fluctuate around a probability of the model. For a uniform point in the square $[0,1]^2$, the proportion satisfying $y \leq x^2$ estimates the area under the parabola; the proportion satisfying $x^2 + y^2 \leq 1$, multiplied by 4, estimates $\pi$. Repetitions must use new draws.

## Simulating and discussing the model

Among 10000 points, if 7840 lie in the quarter disc, the estimate of $\pi$ is $\dfrac{4 \times 7840}{10000} = 3.136$.
Two trials of the same size generally give different values.

## Simulating and discussing the model

Fixing a seed makes a trial reproducible; it neither removes the randomness of the model nor guarantees a specified accuracy.

## Simulating and discussing the model

A simulation of 2000 points gives 675 points below $y = x^2$. What area does it estimate?